

BGIN Meeting Report

Date: TBD

Location: Virtual Meeting

Executive Summary

The meeting focused on the implications of network and asset migration, with a particular emphasis on preserving value during such transitions. The discussion then shifted to the accountable wallet project, which aims to enhance compliance in blockchain transactions through cryptographic technologies. This was followed by an in-depth exploration of privacy pools, their role in improving transaction privacy while maintaining regulatory compliance, and the challenges associated with association set providers. Finally, the meeting concluded with a brief discussion on the sustainability of public goods organizations in the blockchain industry.

Key Discussion Points

- **Network vs. Asset Migration:** The group discussed the complexities of migrating value within different paths for network and asset migrations. Concerns were raised about the potential loss of value when transitioning to a new chain, especially if nodes and developers leave the original chain.
- **Accountable Wallet Project:** This project aims to promote compliant behavior in blockchain transactions by using cryptographic technologies to verify the legitimacy of identities and coin provenance. The protocol is currently under review by IEEE ICBC and has implications for privacy pools on Ethereum.
- **Privacy Pools:** Privacy pools are a solution to enhance transaction privacy on public blockchains like Ethereum. They introduce association sets, which can include compliance checks such as sanctions lists. This ensures that while transactions remain private, they also adhere to regulatory standards.
- **Threat and Harm Database:** The Begin initiative is working on creating a shared global database of threats and harms in the blockchain industry. This aims to improve communication and response mechanisms among various actors in the ecosystem.

Action Items and Next Steps

- **Network Migration Analysis:** Conduct a detailed analysis of the potential risks and benefits associated with network migration, focusing on value preservation.

- **Accountable Wallet Development:** Continue refining the accountable wallet protocol and prepare for its submission to IEEE ICBC. Engage with the Ethereum foundation to explore collaboration opportunities.
- **Privacy Pool Implementation:** Work on implementing association sets within privacy pools to ensure compliance while maintaining user privacy. Collaborate with exchanges and regulatory bodies to establish trusted association set providers.
- **Sustainability of Public Goods Organizations:** Explore different funding models, such as membership associations or subscription-based sponsorship, to sustain public goods organizations in the blockchain industry.

Detailed Session Summary

The meeting began with a discussion on the implications of network and asset migration. Participants highlighted the importance of considering different paths for migrations, especially when dealing with large assets. Concerns were raised about the potential loss of value during transitions to new chains due to the departure of nodes and developers from the original chain.

Next, the conversation shifted to the accountable wallet project. This initiative aims to enhance compliance in blockchain transactions by verifying the legitimacy of identities and coin provenance using cryptographic technologies. The protocol was designed to form a chain of proofs that can be verified by receivers. The project is currently under review by IEEE ICBC and has potential applications for privacy pools on Ethereum.

The discussion then delved into privacy pools, a solution for enhancing transaction privacy while maintaining regulatory compliance. Privacy pools introduce association sets, which include compliance checks such as sanctions lists. This ensures that transactions remain private but also adhere to regulatory standards. The group discussed the implementation of these pools by Oxbow and other organizations, emphasizing the importance of trusted association set providers.

Finally, the meeting concluded with a brief discussion on the sustainability of public goods organizations in the blockchain industry. The Begin initiative is working on creating a shared global database of threats and harms to improve communication and response mechanisms among various actors. Participants were encouraged to explore different funding models, such as membership associations or subscription-based sponsorship, to ensure the long-term sustainability of these initiatives.

Appendix: Full Meeting Transcript (Anonymized)

0.0 --> 3.4: value in the network as opposed to greater value in the asset.
5.1 --> 10.7: So I would consider thinking about what the implications are
10.7 --> 14.1: if you have two different paths for migrations on the two
14.1 --> 16.1: largest assets in there.
16.9 --> 19.8: Because one is a lot simpler and more likely to happen
20.8 --> 23.8: if I migrate value than another, which
23.8 --> 28.2: requires me to delve into the world of legacy migration.
31.1 --> 35.2: Yeah, I get your point.
36.0 --> 39.5: Well, yeah, asset migration and network migration
39.5 --> 41.0: should be different.
41.3 --> 44.3: But at least our scope, especially
44.3 --> 47.5: the competition that's been announced just now,
48.4 --> 50.8: we're focusing on the network migration.
51.9 --> 56.5: And well, even, I don't know, the asset migration,
57.0 --> 58.6: it's just we're talking about the value.
59.6 --> 61.1: There is no guarantee that we can
61.1 --> 63.5: ensure the same value for the same token
65.0 --> 67.0: before migration and after migration
67.0 --> 68.6: if we change the chain.
69.0 --> 72.5: Because anyway, the asset currently, the asset value
72.5 --> 75.0: is determined by the node, the number
75.0 --> 77.2: of nodes operating that chain.
77.6 --> 81.4: That's why Bitcoin Cash is much less than Bitcoin right now.
81.5 --> 86.2: Because developers, miners, nodes, those people
86.2 --> 87.2: all left the chain.
87.2 --> 91.6: And so talking about preserving the asset value,
92.1 --> 96.6: I kind of suspect that moving on to the brand new chain
97.2 --> 100.8: doesn't really guarantee the value preservation.
101.5 --> 102.6: But thank you for asking.
104.7 --> 105.5: Thanks much.
105.5 --> 106.5: That's a very fair point.
107.3 --> 109.1: Future value, I hadn't thought about that.
109.3 --> 109.5: Thanks.
111.2 --> 112.6: Yeah, yeah, I appreciate it.
112.7 --> 113.2: Thanks so much.
118.2 --> 120.8: OK, Keegan, I think you mentioned that you
120.8 --> 123.3: had to drop, unfortunately.
124.8 --> 128.4: Yeah, I got a drop in maybe one or two minutes.
129.3 --> 136.1: OK, well, I'm going to quickly rush to the accountable
wallet
136.1 --> 139.1: section just because this is your research.
139.5 --> 142.9: If you could give us like a quick two minute high level

142.9 --> 146.3: and then I'll continue because I also contributed to this.
147.2 --> 147.9: Yeah, sure.
148.3 --> 151.3: OK, so the accountable, OK, it's kind of a completely different
151.3 --> 152.6: project, but.
152.9 --> 155.4: Yeah, we've moved on from PQC.
155.6 --> 160.2: We're now working on privacy pools and accountable wallet
160.2 --> 161.2: systems.
162.1 --> 167.4: All right, so the accountable wallet project is more
167.4 --> 169.4: about the compliance on blockchain.
169.5 --> 170.3: That's where it began.
170.6 --> 174.3: So the original protocol was proposed and designed
174.3 --> 177.3: by a banker in Japan.
177.8 --> 181.8: And in that, we were trying to mitigate the, I'm sorry,
183.2 --> 187.5: like minimizing the value used for illicit financial activity
187.5 --> 191.6: in P2D blockchain by promoting each user to conduct
192.2 --> 196.5: more compliant behavior and by using
196.5 --> 197.8: some cryptographic technologies.
197.8 --> 204.7: So firstly, the sender of the coin
204.7 --> 210.1: will approve his identity legitimacy and also
210.1 --> 212.4: the coin provenance legitimacy.
213.3 --> 216.4: And we were going to form the chain of proofs.
216.7 --> 221.1: And any receiver can check the legitimacy of the provenance
221.1 --> 227.4: and the identity by verifying the succinct aggregated proof.
227.4 --> 228.7: That's the original idea.
229.1 --> 231.9: But this remained really conceptual.
233.0 --> 237.4: And my work was more simplifying it and also
238.4 --> 240.8: describing the detailed protocol for that.
241.5 --> 246.1: And now it's currently under review of the IEEE ICBC, which
246.1 --> 251.6: is one of the top tier conference for the blockchain.
252.1 --> 259.1: And also, this has some effect on the privacy pools,
259.3 --> 263.4: which is the compliance and privacy preserving protocol
264.0 --> 264.9: working on Ethereum.
265.9 --> 272.1: And simply this compliant behavior
272.1 --> 275.2: and leaving the verifiable proof,
275.8 --> 281.1: such proof can be used as the source of the community
281.1 --> 286.9: curation of the legitimacy or the compliance
286.9 --> 288.3: score for each wallet.
289.0 --> 293.2: And such aggregated curated score
293.2 --> 298.3: can be also used for identifying this wallet is good
298.3 --> 301.2: and this wallet is bad on a community-based decision.

301.8 --> 304.9: And this can support us.
305.1 --> 307.1: This can support the privacy pools operation
308.1 --> 314.0: because this realized the formation of some trustable
314.0 --> 318.4: deposit based on such compliance score,
319.3 --> 326.1: and which is the problem that hasn't been addressed
327.1 --> 330.1: by the original privacy pool proposal.
330.5 --> 335.2: So that's why we are trying to make this protocol
335.2 --> 343.7: as the complement or component, which
343.7 --> 346.8: support the privacy pool and potentially work
346.8 --> 353.3: with the Ethereum foundation and compliance report, how
it work.
354.8 --> 357.0: Yeah, thanks for sharing, Keegan.
357.5 --> 359.0: Really appreciate your time.
360.1 --> 362.1: Yeah, if we can continue, yeah, that'll be appreciated.
362.5 --> 363.1: Thank you so much.
363.2 --> 363.9: Yeah, absolutely.
363.9 --> 367.4: And we'll share the transcript and feedback with you
367.4 --> 368.5: to continue the research.
371.1 --> 373.2: So just to give a bit more, yeah.
374.1 --> 374.6: Thanks, Keegan.
377.7 --> 380.5: A bit more color for those in the room,
380.6 --> 382.7: and we can make this an interactive conversation.
383.3 --> 384.7: Who's aware of privacy pools?
386.8 --> 389.7: OK, I'll explain privacy pools.
389.8 --> 391.9: I think that's relevant background information here.
391.9 --> 395.0: So privacy pools was a paper that was published,
395.5 --> 397.9: I think, in 2003.
400.6 --> 403.0: And Vitalik was also a co-author.
404.9 --> 409.2: It proposes a obfuscation set on chain.
409.7 --> 410.6: So it's similar.
410.8 --> 412.9: It's like an iteration on TornadoCash.
413.2 --> 415.4: If we're aware of TornadoCash, it
415.4 --> 417.7: was the protocol that was sanctioned
417.7 --> 421.3: because DPRK funneled a whole bunch of money laundering
421.3 --> 423.0: through that protocol on Ethereum.
423.7 --> 426.4: So privacy pools is the compliant reaction
427.1 --> 432.1: to improving the TornadoCash outcome, which
432.1 --> 437.0: is to get privacy or anonymity or obfuscation
437.0 --> 440.4: on chain of transactions, which is important.
441.7 --> 445.6: Having private transactions on a public distributed ledger
446.2 --> 449.9: is, in my opinion, one of the most important improvements
449.9 --> 450.6: to Ethereum.
450.6 --> 454.5: Because Ethereum, by default, is this triple entry system that
454.5 --> 457.9: is publicly available, it has been difficult for instit

utions

457.9 --> 461.3: to adopt it because some transactions do
461.3 --> 463.4: need to be kept private and some information
463.4 --> 465.6: about transactions do need to be kept private.
466.2 --> 468.8: So privacy pools is a solution to that.
469.3 --> 472.2: And the key difference between TornadoCash and privacy
pools
472.7 --> 477.3: is privacy pools has this something
477.3 --> 480.8: called an association set, which is
481.8 --> 485.4: like a registry within the pool that
485.4 --> 490.6: can have compliance or regulatory checks and balances i
n place.
491.2 --> 494.7: So one of the first implementers of privacy pools
494.7 --> 500.3: as the paper into real world applications was Oxbow.
501.0 --> 503.9: And their association set, the first one they did,
504.2 --> 505.3: was sanctions.
505.3 --> 508.4: So all of the international sanctions,
508.9 --> 512.6: all the wallets that have been listed by different law
512.6 --> 515.4: enforcement government bodies that we should not
515.4 --> 519.0: interact with on chain, are now existing
519.0 --> 521.4: within what's called an association
521.4 --> 523.3: set within the privacy pool.
523.9 --> 528.2: So you can receive a credential when you make
528.2 --> 530.1: a deposit into a privacy pool.
530.2 --> 533.6: So say, for example, I deposit 1 eth into a privacy poo
l
533.6 --> 536.0: and I'd like to get 1 eth out the other side.
536.5 --> 538.5: The idea behind a privacy pool is
539.1 --> 542.0: if there is a certain sort of threshold,
542.1 --> 544.4: so 100 people all deposit the same amount,
544.6 --> 546.7: so everyone deposits 1 eth into the pool,
547.3 --> 549.4: then it all mixes up in the middle
550.0 --> 552.2: and you can withdraw 1 eth.
552.5 --> 558.7: And that obfuscates your flow of the asset from one ini
tial
558.7 --> 562.4: address that you send the money in with to the address
562.4 --> 563.9: that you withdraw it from.
564.3 --> 569.2: So it breaks the chain of, I guess, transaction monitor
ing.
570.1 --> 573.9: However, while this sounds like a really cool technolog
y,
574.2 --> 576.9: it does fly in the face of compliance.
577.2 --> 579.4: So these association set providers,
579.7 --> 585.1: all of the verifiable records within these pools
586.0 --> 588.3: need to have certain registries and checks.
589.3 --> 592.8: So what Begin is working on is outlining
592.8 --> 596.5: what some of these association set providers,

598.7 --> 602.0: how they're constructed, and how they're also governed.
602.3 --> 605.4: Because you could have a nefarious association set
605.4 --> 608.6: provider that then purposely tries to create compliance
608.6 --> 609.8: within the pool.
610.2 --> 612.2: So then at one point in the future,
612.7 --> 615.1: a nefarious actor can funnel all of their money
615.1 --> 617.0: through the privacy pool.
617.0 --> 620.9: So this improvement that we've proposed
622.2 --> 626.3: is like inclusion of a verifiable compliance
626.3 --> 629.8: record into the association set providers within the po
ol.
631.1 --> 633.4: So the first example is the sanctions.
633.7 --> 637.3: But you could imagine that this could be a specific exc
hanges
637.3 --> 643.9: KYC network that they then put as an association set
643.9 --> 646.5: registry within a particular privacy pool
646.5 --> 647.3: that they're deploying.
647.7 --> 650.4: So then their users can get financial privacy
650.4 --> 652.6: on the public permissionless blockchain
654.4 --> 658.1: without revealing where the money comes from, et cetera.
659.8 --> 660.3: You have a question?
665.9 --> 667.8: OK, I can continue.
675.6 --> 676.6: Yeah, it's fungible.
678.3 --> 681.9: Yeah, so traditional finance does this anyway
681.9 --> 683.1: because it's centralized, right?
684.4 --> 686.9: So traditional finance doesn't really have this problem.
687.0 --> 689.9: You could imagine a privacy pool is actually a bank, ri
ght,
690.0 --> 691.9: if you think about it that way.
692.0 --> 693.7: So you put a whole bunch of money into a bank.
693.8 --> 694.8: It all mixes up.
694.9 --> 695.4: It's fungible.
695.6 --> 697.8: The \$10 that I put in could be the \$10 that I put out.
697.8 --> 699.2: The \$10 that you take out doesn't matter.
699.3 --> 700.2: The ledger is updated.
712.5 --> 716.9: Yeah, so the sanctions is kind of just
716.9 --> 718.8: the first expression of this.
719.0 --> 721.3: And it was because the industry was
721.3 --> 726.1: under a lot of scrutiny for the DPRK hacks, et cetera.
727.9 --> 730.4: I don't want to call upon Arianna,
730.5 --> 733.6: but she's an expert in the room on that.
735.4 --> 739.1: So that was just the industry's first expression of,
739.2 --> 742.5: we need to make sure that if we're offering privacy
742.5 --> 744.6: preserving technology, such as privacy pools,
745.2 --> 750.3: we need to have this initial cutoff
750.3 --> 753.0: of nefarious actors that are publicly disclosed
753.0 --> 753.8: nefarious actors.

753.9 --> 757.7: And sanctions lists are a pretty clear indicator of that.
758.9 --> 761.6: But the purpose and, I think, the ambition
761.6 --> 765.9: of the privacy pools paper is to be much greater than that
765.9 --> 769.7: and for there to be privacy pools for these intranetworks
769.7 --> 772.0: within the public distributed ledger.
772.2 --> 775.0: So if a bank decides to deploy a real world
775.0 --> 779.1: asset on a public permissionless chain like Ethereum
779.1 --> 782.2: using its stable coins, but that real world asset
782.2 --> 786.4: needs some sort of compliance, wholesale investor certificate,
786.6 --> 790.5: et cetera, an association set is that compliance governance
790.5 --> 793.6: that would sit within the pool.
794.3 --> 796.6: So everyone who wants to buy that real world asset
796.6 --> 800.0: puts the certain amount of value that
800.0 --> 803.9: needs to be used to purchase that real world
803.9 --> 805.2: asset into the pool.
805.2 --> 808.4: And then they can receive the asset out the other end,
809.4 --> 812.8: either for the same amount of value or for plus minus,
813.0 --> 814.9: depending on what happens within the pool.
815.4 --> 818.5: But within the pool, you have those checks and balances
818.5 --> 819.2: in place.
819.9 --> 823.7: And then because it is happening within that intra ecosystem,
824.1 --> 825.8: you get these privacy guarantees.
825.8 --> 831.5: So you don't have to do on-chain proofs of personally
831.5 --> 833.2: identifiable data all the time.
833.2 --> 837.8: It can actually sit within just the association set providers
837.8 --> 838.6: governance.
840.5 --> 842.9: And the interesting thing about that
842.9 --> 846.7: is you then potentially get these cross privacy pools
846.7 --> 850.3: association sets where, for example, there would be like
850.3 --> 852.6: one bank would have a association set
852.6 --> 854.9: and the other bank would have another association set.
855.1 --> 857.2: And if I wanted to transfer between these banks,
858.7 --> 862.3: we could check that I'm on both of these registries.
863.3 --> 865.8: And only everyone else who is also
865.8 --> 869.4: on both of these registries are the deposits and withdrawals
869.4 --> 871.9: that I can mix or interact with.
879.2 --> 883.0: So that progresses to one of the other initiatives
883.0 --> 888.2: that Begin is looking to provide to the industry.

888.7 --> 891.2: And that's compressing a lot of the threats and harms
891.2 --> 893.9: into equations or math.
894.6 --> 897.5: And then for us to then, depending on the context,
898.3 --> 900.4: so for example, if we're protecting from sanctions
900.4 --> 903.7: or if we're protecting from hackers, to expand that out.
904.5 --> 913.0: So open to participants to adding their own threats
913.0 --> 915.4: and harms to this common lexicon.
915.9 --> 918.4: But we'll be hosting, thanks for having us here,
919.4 --> 923.2: we'll be hosting or providing governance
923.2 --> 927.8: for a shared global database of threats and harms
927.8 --> 929.2: in the blockchain industry.
930.2 --> 931.9: So then there can be better communication
931.9 --> 935.4: between these sort of actors.
935.7 --> 939.3: So SEAL 901 does the security assistance and the incident
939.3 --> 940.0: response.
941.3 --> 943.6: MITRE provides the machine readable
943.6 --> 946.6: and the technical understanding of both defend and attack
946.6 --> 948.9: techniques in cybersecurity.
949.7 --> 953.5: So this gets to, as I started with the Begin conversation,
953.8 --> 955.8: we have these three core working groups.
956.0 --> 957.5: This is the cyber working groups.
958.3 --> 963.1: Focus is building this AI machine readable,
963.3 --> 967.5: but also human reconstructable database of threats and harms
968.3 --> 971.1: to build a more efficient communication protocol.
971.8 --> 973.9: So different actors across the world
973.9 --> 976.8: can respond when an exchange gets hacked
976.8 --> 982.1: or some sort of threat or harm exists
982.1 --> 984.4: in the public permissionless blockchain space.
991.5 --> 993.2: We're discussing that.
994.3 --> 999.2: It's mostly in kind through sponsorship for our events
999.2 --> 1002.8: traditionally, but we're looking for different ways
1002.8 --> 1007.3: to gain sustaining sponsors who are more interested
1007.3 --> 1008.9: in the outcomes and the technologies
1008.9 --> 1012.8: that we create in more of a membership association
1012.8 --> 1013.9: subscription model.
1014.6 --> 1017.1: And if you have interested parties in that,
1017.2 --> 1017.5: I would encourage you to reach out to them.
1017.5 --> 1020.0: Love to have that conversation because a lot of the work
1020.0 --> 1022.4: that we do is right now public goods
1022.9 --> 1027.4: and it is difficult to run public goods organizations.
1027.9 --> 1031.2: And yeah, it's from the good of our heart,
1031.2 --> 1032.3: but that's not sustainable.

1032.8 --> 1035.9: So, yeah, happy to have that conversation in more detail.

1037.1 --> 1041.1: Good, you have a big heart, but it's not fair.

1042.1 --> 1042.9: Yeah, exactly.

1044.1 --> 1046.3: Some of the other research that we're working on

1046.3 --> 1049.7: is this distinguishing between the two domains.

1049.8 --> 1053.8: And this was Amanda's and my work

1054.7 --> 1057.5: when she is involved with Begin.

1057.7 --> 1060.6: And a lot of issues that occur in the courts

1060.6 --> 1062.9: is there's a difference between blockchain analytics

1062.9 --> 1064.3: and blockchain forensics.

1064.5 --> 1067.1: And that's the difference between evidence

1067.1 --> 1073.1: and analysis of information that exists on chain, right?

1073.5 --> 1076.2: So analytics is like the firms like Elliptic

1076.2 --> 1077.8: or Chainalysis, et cetera.

1078.2 --> 1080.9: But then how does that information translate

1080.9 --> 1083.7: into actually being legally enforceable

1084.6 --> 1088.8: and legally verifiable in court or understood by judges?

1089.6 --> 1093.3: So this is another outcome that we're looking to generate

1093.3 --> 1096.3: at Begin through this common lexicon

1096.3 --> 1100.7: is identifying what it takes for blockchain information

1100.7 --> 1103.0: to be forensic versus analytic,

1104.1 --> 1106.0: which has been a common question

1106.0 --> 1110.8: by a lot of the legal folk that join our conversations.

1115.0 --> 1120.0: So here's kind of the breakdown of an hour approach

1120.6 --> 1125.1: to finding a solution to generating this common lexicon.

1126.9 --> 1128.5: So there's harmful activities,

1128.7 --> 1133.0: tracing, de-anonymization or de-pseudo-anonymization,

1133.4 --> 1135.6: and then the wallet clustering work

1135.6 --> 1138.5: that a lot of these analytic firms do.

1145.5 --> 1150.9: Is there a similar service for chains

1150.9 --> 1155.0: or for looking at blockchain changes over time?

1156.3 --> 1156.6: Yes.

1160.7 --> 1163.2: Well, like a blockchain,

1163.8 --> 1165.9: all of the transactions that fit within a block

1165.9 --> 1168.6: are always constant.

1169.0 --> 1171.7: It's a deviation chain.

1171.9 --> 1173.2: So you can always go back

1173.9 --> 1175.4: and you can see the full provenance,

1175.5 --> 1177.9: which is one of its strengths,

1178.1 --> 1180.2: but also in a way it's curse

1180.9 --> 1183.7: because the more compute you throw at it,
1184.2 --> 1187.6: the more knowledge outside of the system
1187.6 --> 1190.5: that you attach to knowledge within the blockchain system,
1191.6 --> 1197.3: the less privacy, security focused it is
1197.3 --> 1198.3: for the people who interact
1198.3 --> 1200.0: on the public permission blockchains.
1200.2 --> 1202.4: So the assumption is that over time
1202.4 --> 1204.8: all of these fully transparent chains
1204.8 --> 1206.9: are going to reveal way more information
1207.5 --> 1209.9: about the transactions that happened 10 years ago.
1210.2 --> 1211.7: So all the guys and girls
1211.7 --> 1214.3: that were on the dark markets using Bitcoin,
1215.0 --> 1216.8: like in many, many years time,
1216.9 --> 1218.5: there's going to be enough compute
1218.5 --> 1220.1: thrown at this information and data
1220.1 --> 1223.7: for a lot of that to almost get caught up
1223.7 --> 1225.2: with more knowledge.
1226.5 --> 1228.6: So that's in part, yeah,
1228.7 --> 1230.1: the race that we're trying to make.
1230.4 --> 1232.9: And it's obviously always a balance
1232.9 --> 1236.8: that needs to be struck between that paradox.
1242.1 --> 1244.4: Again, I can, with everyone in the group,
1244.5 --> 1247.6: I'll probably send an email with the presentation
1247.6 --> 1250.3: and you can find on the website,
1250.5 --> 1256.7: I think here, begin-global.org is our homepage
1258.0 --> 1259.7: to interact with the community.
1260.1 --> 1262.0: We have what's called a discourse group,
1262.3 --> 1264.2: which is just like a forum page.
1265.4 --> 1268.6: And then this is my specific research
1268.6 --> 1270.0: or the living documentation.
1271.5 --> 1274.9: But I also have a website called agentprivacy.ai
1275.5 --> 1277.9: and spellweb.ai is that visualization.
1277.9 --> 1282.6: So you can actually observe that knowledge substrate
1282.6 --> 1284.0: that I'm offering to the world.
1285.9 --> 1287.8: So I'm going to conclude here.
1288.0 --> 1289.7: Any final questions are welcome.
1290.5 --> 1292.6: We have the space until one,
1292.7 --> 1296.5: but the end of talks is at 1230.
1296.9 --> 1298.4: And yeah, thank you so much
1298.4 --> 1300.6: for being a part of this conversation.
1302.8 --> 1304.2: Yeah.
1306.0 --> 1308.8: Yeah, end.
1341.4 --> 1342.0: Thank you.
1342.2 --> 1343.6: I would very much appreciate that.
1343.9 --> 1346.3: It's, it can only be a solar
1346.3 --> 1348.4: entrepreneur research for so long,

1349.4 --> 1352.6: but I feel like it tends to be a,
1352.7 --> 1354.7: I've seen this happen a lot more.
1355.5 --> 1356.9: And I just want you to feel free to reach out.
1357.5 --> 1359.0: You have two minutes.
1359.7 --> 1359.9: Yeah.
1362.2 --> 1365.8: It's a trend that I'm noticing among a lot of people
1365.8 --> 1367.5: in the industry at the moment
1367.5 --> 1370.6: where people are no longer being employed
1370.6 --> 1374.8: and everyone's running it back with AI and agents
1374.8 --> 1377.0: and creating all of these things very quickly,
1377.5 --> 1378.9: but none of them have revenue.
1379.3 --> 1382.2: So I'm in the same boat where I'm able to create
1382.2 --> 1386.2: all of this like expressions of the research very eas
ily,
1386.7 --> 1388.8: but then finding a way to have that funded
1388.8 --> 1390.7: is another question altogether.
1391.8 --> 1393.7: So yeah, happy to have that conversation.
1393.7 --> 1394.2: I would love.
1396.4 --> 1396.9: Yeah.
1420.0 --> 1421.4: Yeah.
1446.8 --> 1447.4: Yeah.
1457.0 --> 1457.3: Oh yeah.
1457.6 --> 1457.9: It's a trend.
1458.4 --> 1458.7: Thank you.
1462.6 --> 1463.5: Oh, that's good.
1464.2 --> 1464.5: Good.
1466.1 --> 1467.0: I'll do that.
1467.4 --> 1467.8: It's good.
1468.0 --> 1468.0: That's good.
1479.3 --> 1480.7: Yeah.
1501.8 --> 1503.2: Yeah.
1531.2 --> 1532.6: Yeah.
1559.5 --> 1560.9: Yeah.
1588.0 --> 1588.2: Yeah.
1602.0 --> 1604.8: It's great.
1604.8 --> 1607.5: So it's a simple thing, all of that.
1608.6 --> 1611.8: It's just offering and...
1616.3 --> 1616.8: Yeah.
1620.5 --> 1621.5: Oh, very good.
1621.9 --> 1622.2: Thank you.
1622.4 --> 1622.5: Nice.
1622.8 --> 1625.1: But it's really like, part of-
1625.1 --> 1625.7: Lots.
1628.7 --> 1629.7: Okay, great.
1629.9 --> 1630.7: I'm glad you're here.
1635.6 --> 1641.1: So, you know what happened, the concept was that it w
as a strong point to start with,

1641.3 --> 1643.8: but I thought, you know, this is the thing, right?
1644.3 --> 1646.5: So, I don't have to get rid of it.
1647.3 --> 1648.0: This is it.
1648.4 --> 1649.4: Get rid of it.
1649.4 --> 1649.6: Get rid of it.
1650.0 --> 1651.5: This is the first step.
1651.7 --> 1651.8: That's it.
1652.1 --> 1653.5: And I think now I've improved.
1655.9 --> 1657.2: I have improved.
1657.7 --> 1657.8: That's great.
1748.6 --> 1750.2: Thank you.
1770.6 --> 1773.4: Thank you.
1790.5 --> 1793.3: Thank you.
1843.6 --> 1845.0: Thank you.
1857.7 --> 1858.3: Okay.
1858.4 --> 1858.4: Okay.
1858.4 --> 1858.4: Okay.
1858.4 --> 1859.1: Okay.
1868.6 --> 1869.3: Okay.